

SOLID MECHANICS

IV

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22914927

MODULE	SME3701 — Solid Mechanics IV
ASSIGNMENT	Assignment 1 of 2026
DUE DATE	30 April 2026 at 22:00
Github Link	https://github.com/PrideKG29/22914927.SME3701A1.git



```
1 %% 1.1 - Natural Angular Frequency
2 % 22914927
3
4 % System parameters
5 m = 5;          % mass [kg]
6 k = 2000;       % spring stiffness [N/m]
7
8 % Natural angular frequency [rad/s]
9 wn = sqrt(k/m);
10
11 % Solution
12 fprintf('Natural Angular Frequency: wn = %.4f rad/s\n', wn);
13 QuestionOneCode
14 Natural Angular Frequency: wn = 20.0000 rad/s
```

```

1 %% 1.2 - Displacement x(t)
2 % 22914927
3
4 % System parameters
5 m = 5;          % mass [kg]
6 k = 2000;       % spring stiffness [N/m]
7 x0 = 0.02;      % initial displacement [m]
8
9 % Natural angular frequency [rad/s]
10 wn = sqrt(k/m);
11
12 % Time vector
13 t = linspace(0, 5, 1000); % 0 to 5 seconds, 1000 points
14
15 % Displacement equation:  $x(t) = x_0 \cos(\omega_n t)$ 
16 x = x0 * cos(wn * t);
17
18 % Solution
19 fprintf('Displacement computed:  $x(t) = 0.0200 \cos(20.0000 t)$ \n', x0, wn);
20 fprintf('At t = 0 s : x = 0.0200 m\n', x(1));
21 fprintf('At t = 5 s : x = 0.0172 m\n', x(end));
22 QuestionTwo
23 Displacement computed:  $x(t) = 0.0200 \cos(20.0000 t)$ 
24 At t = 0 s : x = 0.0200 m
25 At t = 5 s : x = 0.0172 m

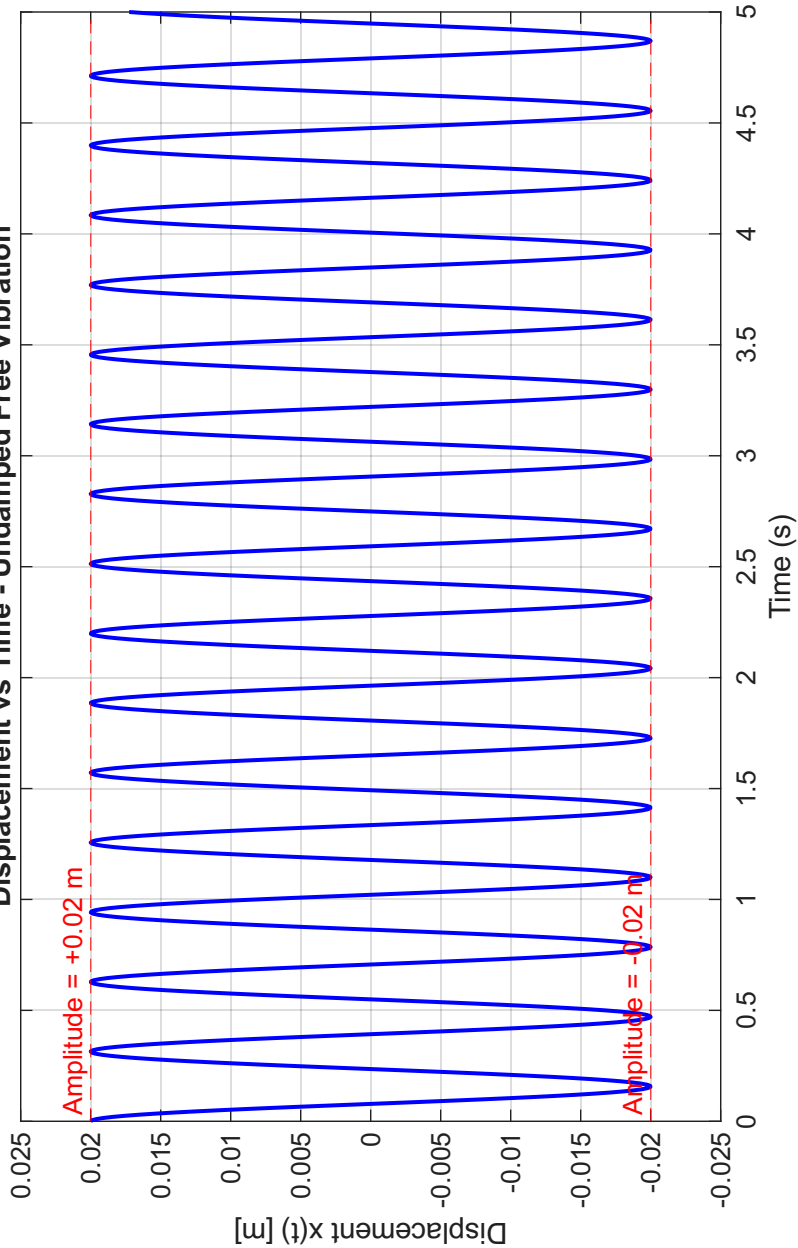
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```

1 %% Q1.3 - Plot Displacement vs Time
2 % 22914927
3
4 % System parameters
5 m = 5;      % mass [kg]
6 k = 2000;   % spring stiffness [N/m]
7 x0 = 0.02;  % initial displacement [m]
8
9 % Natural angular frequency [rad/s]
10 wn = sqrt(k/m);
11
12 % Time vector
13 t = linspace(0, 5, 1000); % 0 to 5 seconds, 1000 points
14
15 % Displacement:  $x(t) = x_0 \cos(\omega_n t)$ 
16 x = x0 * cos(wn * t);
17
18 % Solution
19 figure;
20 plot(t, x, 'b', 'LineWidth', 1.5);
21
22 % Labels and formatting
23 title('Displacement vs Time - Undamped Free Vibration');
24 xlabel('Time (s)');
25 ylabel('Displacement x(t) [m]');
26 grid on;
27 xlim([0 5]);
28 ylim([-0.025 0.025]);
29
30 % Annotate key values
31 yline(x0, '--r', 'Amplitude = +0.02 m', 'LabelHorizontalAlignment', 'left');
32 yline(-x0, '--r', 'Amplitude = -0.02 m', 'LabelHorizontalAlignment', 'left');
33 QuestionThreeGraphicalSolution

```

Displacement vs Time - Undamped Free Vibration

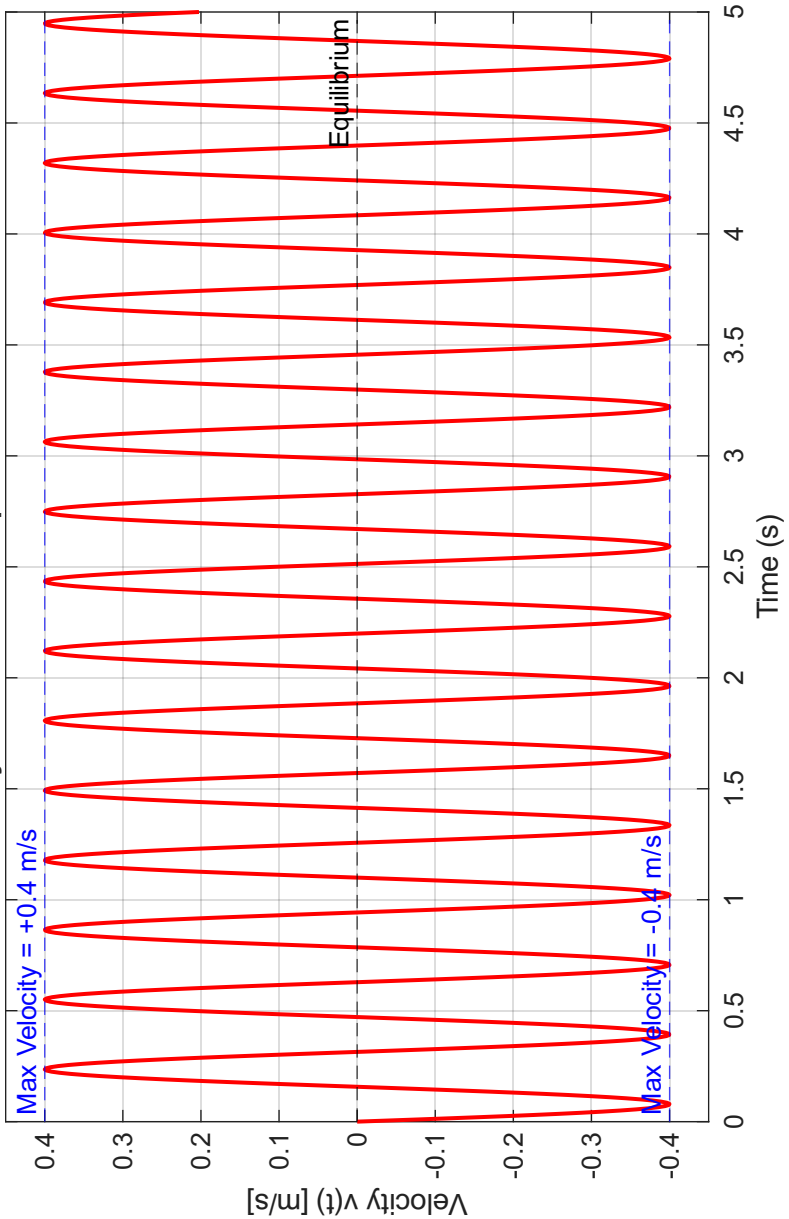


```

1 %% 1.4 - Velocity v(t) and Plot
2 % 22914927
3
4 % System parameters
5 m = 5;          % mass [kg]
6 k = 2000;       % spring stiffness [N/m]
7 x0 = 0.02;      % initial displacement [m]
8
9 % Natural angular frequency [rad/s]
10 wn = sqrt(k/m);
11
12 % Time vector
13 t = linspace(0, 5, 1000); % 0 to 5 seconds, 1000 points
14
15 % Displacement and Velocity
16 x = x0 * cos(wn * t);      % x(t) = x0*cos(wn*t) Displacement
17 v = -x0 * wn * sin(wn * t); % v(t) = -x0*wn*sin(wn*t) velocity
18
19 % Solution
20 fprintf('Velocity equation: v(t) = -.4f * sin(.4f * t)\n', x0*wn, wn);
21 fprintf('Max velocity amplitude : .4f m/s\n', x0*wn);
22 fprintf('At t = 0 s : v = .4f m/s\n', v(1));
23
24 % Plot velocity
25 figure;
26 plot(t, v, 'r', 'LineWidth', 1.5);
27
28 % Labels and formatting
29 title('Velocity vs Time - Undamped Free Vibration');
30 xlabel('Time (s)');
31 ylabel('Velocity v(t) [m/s]');
32 grid on;
33 xlim([0 5]); % x-axis limits
34 ylim([-0.45 0.45]); %Y-axis limits
35
36 % Annotate key values
37 yline( x0*wn, '--b', 'Max Velocity = +0.4 m/s', 'LabelHorizontalAlignment', 'left');
38 yline(-x0*wn, '--b', 'Max Velocity = -0.4 m/s', 'LabelHorizontalAlignment', 'left');
39 QuestionfourCode
40 Velocity equation: v(t) = -0.4000 * sin(20.0000 * t)
41 Max velocity amplitude : 0.4000 m/s
42 At t = 0 s : v = -0.0000 m/s
43
44
45

```

Velocity vs Time - Undamped Free Vibration



1.5. Comments

- **Nature of the Motion** : The system is undergoing a Simple harmonic motion, oscillating at a constant amplitude of ± 0.02 m with a period of $T = 0.314$.
- **Effect of Zero damping** : No energy is dissipated, the amplitude remains constant at 0.02 , ($C=0$). The system never goes to rest. Mechanical energy is conserved
- **Displacement and Velocity Relationship** : The two quantities are always 90° apart. When the mass is at its furthest point, it stops momentarily before coming back ,so velocity is zero. When it passes through the middle, it is moving fastest at 0.4 m/s.

Grithub link: <https://github.com/PrideKG29/22914927.SME3701A1.git>

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